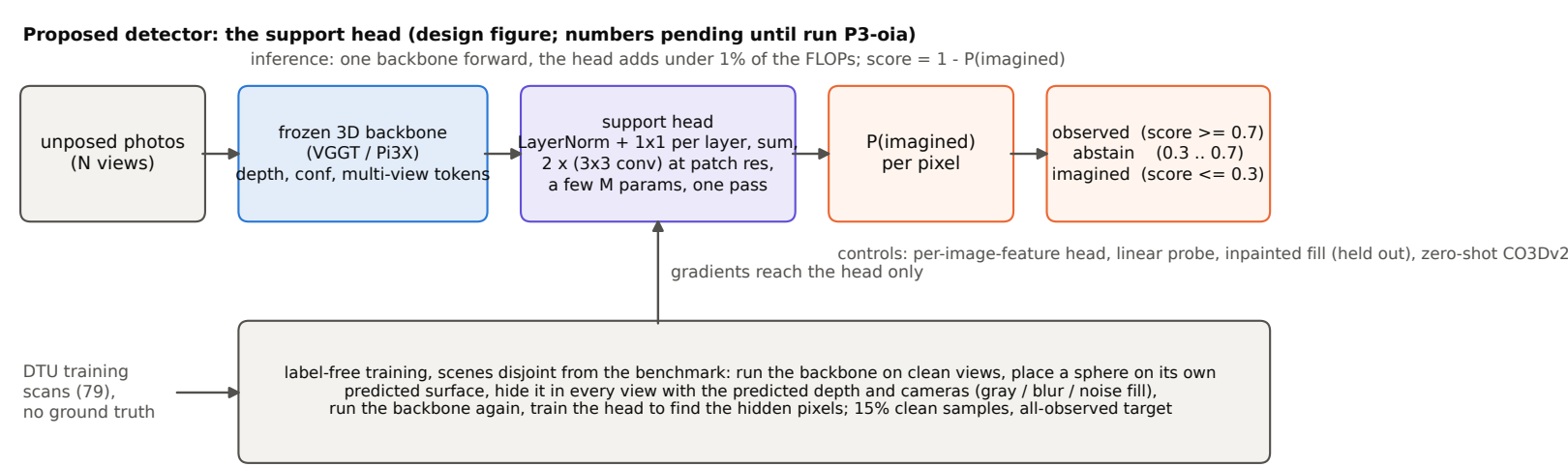
Proposed detector: the support head (design figure; numbers pending until run P3-oia)
inference: one backbone forward, the head adds under 1% of the FLOPs; score = 1 - P(imagined)
unposed photos (N views)
frozen 3D backbone (VGGT / Pi3X) depth, conf, multi-view tokens
support head LayerNorm + 1x1 per layer, sum, 2 x (3x3 conv) at patch res, a few M params, one pass
P(imagined) per pixel
observed (score >= 0.7)
abstain (0.3 .. 0.7)
imagined (score <= 0.3)
controls: per-image-feature head, linear probe, inpainted fill (held out), zero-shot CO3Dv2
gradients reach the head only
DTU training scans (79), no ground truth
label-free training, scenes disjoint from the benchmark: run the backbone on clean views, place a sphere on its own predicted surface, hide it in every view with the predicted depth and cameras (gray / blur / noise fill), run the backbone again, train the head to find the hidden pixels; 15% clean samples, all-observed target